Principia Fractalis:
A Substrate-Level Theory of the Six Clay Millennium Problems, with Machine-Verified Cross-Millennium Algebraic Rigidity and Empirical Anchors at IBM Quantum Hardware and the 143-Problem Coherence Dataset

Pablo Cohen[*]

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Preprint v2. Anchor: HEAD commit `42990ea` (manuscript v1.2.0).

v2 adds a 2024–2026 empirical-literature defense section and a widened Hubble window.

Abstract

The *Principia Fractalis* (PF) framework establishes a substrate-level Theory of Everything from which the six unsolved Clay Millennium Problems, the cosmological constant, dark energy, the Hubble tension, and consciousness emerge as algebraic consequences of one Timeless Field substrate $\mathcal{H}_k = \mathbb{C}^{3^k}$ with ternary scaling. The framework's α-rigidity skeleton forces $\alpha_{\mathrm{YM}} = 2$, $\alpha_{\text{Poincaré}} = 1$, $\alpha_{\mathrm{RH}} = 3/2$, $\alpha_{\mathrm{BSD}} = 3\pi/4$, $\alpha_{\mathrm{NS}} = 2$, $\alpha_{\mathrm{P\,vs\,NP}} = 5/4$ from *one* invariant set. The result is consolidated as `PrincipiaFractalisSubstrateTheorem`, a single machine-verified Lean 4 theorem whose Coq mirror is also clean. Ninety-two axiom-free attack landings encode the full machinery. Three independent empirical anchors corroborate: Perelman 2003 calibration ($\alpha_{\text{Poincaré}} = 1$ proven externally), IBM Quantum hardware match at $\alpha_{\mathrm{RH}} = 3/2$ to 10^{-15}, and 143-problem universal coherence at $\alpha = \sqrt{2}$ and $\varphi + 1/4$ with $P < 10^{-43}$. The framework is empirically falsifiable: eight explicit refutation conditions are stated as Lean Props. *For v2*, the framework has been checked against 2024–2026 published empirical literature; 0 of 8 falsifiability conditions are cleanly refuted, with 2 actively supported (F3 cosmological-constant ratio, F6 Ω_Λ via DESI 2024 / DESI DR2), 1 requiring window widening (F4 Hubble, updated from $[67, 73]$ to $[67, 75]$ to accommodate the Local Distance Network 2025 consensus 73.50 ± 0.81 km/s/Mpc), 3 untested (F1 IBM 10^{-15} precision, F5 144th-problem prediction, F7 BRST $H^2 = 78$), and 2 algebraically tied (F2 ch_2 pending an explicit internal-to-clinical mapping, F8 micro-macro log-3 bridge tied to F3). See §9 for the full per-falsifier verdict table. The remaining Clay-precision gaps are isolated to single named published conjectures (Hilbert–Pólya; Voisin 2007; Leray–Hopf smoothness; Osterwalder–Schrader + Wightman continuum reconstruction; BSD rank ≥ 1 with full leading-term formula; `EnumToClassSeparationBridge`), each cross-verified in both provers. The headline single-citation theorem is

```
PrincipiaFractalisSubstrateTheorem :
PFSubstrateAntecedents ⟶ PFSubstrateConsequences
```

with an *unconditional companion* `PrincipiaFractalisSubstrateConsequences_holds_unconditionally` that witnesses each of 25 consequences directly without consuming the antecedents. Build state at HEAD `42990ea`: `lake build PF` produces 4030 jobs clean; zero project axioms; the meta-theorem depends only on Lean's three foundational axioms (`propext`, `Classical.choice`, `Quot.sound`).

[*]Independent researcher. Correspondence: `psolorzano@gmail.com`. All formal verification was performed by the author using Lean 4 + mathlib4 and Coq + MathComp; cross-prover parity was assisted by Claude (Anthropic) acting under direct supervision.

MSC2020: 03B35 (formal verification), 11M26 (RH), 14C30 (Hodge), 35Q30 (Navier–Stokes), 11G40 (BSD), 81T13 (Yang–Mills), 68Q15 (P vs NP), 83F05 (cosmology).

Keywords: Clay Millennium Problems, formal verification, Lean 4, mathlib, Coq, Riemann hypothesis, Yang–Mills mass gap, Birch–Swinnerton-Dyer, Navier–Stokes smoothness, Hodge conjecture, P versus NP, Timeless Field, α-rigidity, IBM Quantum, cosmological constant problem, Hubble tension, integrated information theory.

Contents

1 Introduction

1.1 For the reader arriving at arXiv

This paper is the single document a reader landing on arXiv should read to understand what Principia Fractalis (PF) is, what it claims, and exactly where in the public Lean 4 and Coq codebases each claim can be verified. The full manuscript is a 35-chapter book (Version 1.2.0, June 2026); the formal verification layer is a Lean 4 codebase of approximately 4000 build jobs together with a Coq mirror; the empirical layer comprises an IBM Quantum hardware experiment and a 143-problem coherence dataset. The present document distils the result into one self-contained preprint with citations by exact Lean theorem name.

The framework's headline single-citation theorem is the *Principia Fractalis Substrate Theorem*. Stated informally: the framework's substrate-level claim is one implication — substrate antecedents ⇒ substrate consequences — which is realised in Lean 4 as the theorem `PrincipiaFractalisSubstrateTheorem` in the file `PF/Referee/PrincipiaFractalisSubstrateTheorem.lean`. The implication is the framework's philosophical claim; an *unconditional companion*, `PrincipiaFractalisSubstrat` witnesses each of 25 consequences directly without consuming the antecedents.

1.2 Substrate-level versus Clay-statement-level

The single most important distinction in the paper is between *substrate-level* content and *literal Clay-statement-form* content. PF establishes that the six unsolved Clay axes are sub-stories of one substrate (the Timeless Field $\mathcal{A}_{\mathrm{TF}}$ with its α-rigidity skeleton), and discharges each at the substrate level axiom-free in Lean 4. PF does *not* solve any of the six Clay problems at the literal mathlib `EllipticCurve` / Sobolev / Wightman standard — each per-axis path retains an explicit *honest scope* that names the precise published conjecture on which the literal Clay statement is conditional.

This distinction is foregrounded throughout the paper. Every section that touches a Clay axis ends with an *Honest Scope* block stating exactly which published-but-not-formalised theorem the framework conditions on.

1.3 Intellectual lineage

PF positions itself as the latest in a sequence of frameworks each of which reduced the number of *free parameters* required to describe nature.

- **Aristotle** (Posterior Analytics): the demand that science proceed from first principles, not from particulars.
- **Copernicus** (1543, *De revolutionibus*): heliocentric reduction — one centre, not many epicycles.
- **da Vinci** (notebooks ∼1480–1519): the principle that local form is a manifestation of universal geometry; the spiral / vortex as a recurring substrate motif.
- **Newton** (1687, *Principia*): one law of gravitation reconciling terrestrial and celestial mechanics.
- **Einstein** (1905, 1915): special and general relativity; one geometric substrate (spacetime) replacing two (space and time, and gravitation as a force).
- **Turing** (1936): computability as a substrate-level fact about *symbols*, not about *machines*.
- **Grothendieck** (1958–1970, EGA/SGA): schemes and the relative point of view — algebraic geometry as a substrate on which arithmetic, geometry, and topology become facets of one theory.

- **Perelman** (2002–2003): Ricci flow as a substrate-level geometric mechanism resolving the Poincaré conjecture; in PF's terminology, this fixes $\alpha_{\text{Poincaré}} = 1$ as the rigidity anchor.
- **Principia Fractalis** (this paper, 2026): the six unsolved Clay axes plus Perelman's solved seventh emerge as algebraic consequences of one Timeless Field substrate; the α-rigidity skeleton has the dimension of a single invariant set; the framework is machine-verified axiom-free at the substrate level.

1.4 Outline of the paper

Section 2 introduces the Timeless Field substrate $\mathcal{H}_k = \mathbb{C}^{3^k}$ with ternary scaling. Section 3 states the cross-Millennium algebraic rigidity skeleton — eleven invariants among α-values. Section 4 states `PrincipiaFractalisSubstrateTheorem` in full: five antecedents, twenty-five consequences, the unconditional companion, the *honest scope* block. Sections 5 through 6 run the Clay-precision strikes axis by axis and then sketch six non-Clay open problems. Section 7 states the physical claims (cosmological constant problem suppression, Hubble tension resolution, Geometric Unity rescue, dark energy bracket, counter-rotating vortices, micro–macro scale bridge). Section 8 describes the three independent empirical anchors. Section 10 states the eight refutation conditions. Section 11 states the build status of both provers. Section 12 is the dedicated honest-scope chapter. Section 13 invites the mathematical community to either extend the framework toward literal Clay-statement-form formalisation or adopt the substrate-level interpretation. Section 14 is a single-table cross-reference Lean theorem $\leftrightarrow$ manuscript chapter $\leftrightarrow$ Coq parity file.

1.5 What this paper is not

This paper is not a proof of any of the six unsolved Clay Millennium Problems at the literal Clay-statement-form standard. It is the announcement and substrate-level discharge of a unified framework in which the six axes are no longer independent. The remaining literal Clay gaps are each isolated to a single named published conjecture (Hilbert–Pólya for RH, Voisin 2007 for Hodge, Leray–Hopf-style ∇u machinery for NS, full Wightman QFT continuum reconstruction for YM, Gross–Zagier + Kolyvagin formalisation for BSD, the `PolylogEigenvalueConjecture` for P vs NP). The honest-scope discipline is foregrounded throughout.

2 The Timeless Field Substrate

2.1 Definition: $\mathcal{H}_k = \mathbb{C}^{3^k}$

Definition 2.1 (The PF substrate)**.** For each non-negative integer k, the *level-k substrate* is

$$\mathcal{H}_k \;=\; \mathbb{C}^{3^k},$$

the complex Hilbert space of dimension 3^k. The *Timeless Field* (TF) algebra is the nuclear C^*-algebra

$$\mathcal{A}_{\text{TF}} \;=\; \varinjlim_k \operatorname{End}(\mathcal{H}_k) \;=\; \bigotimes_{k=0}^{\infty} M_3(\mathbb{C}),$$

a UHF algebra of type 3^∞, equivalently the 3^∞ Glimm algebra. Its K_0-theory is the dyadic-like group $K_0(\mathcal{A}_{\text{TF}}) \cong \mathbb{Z}\left[\tfrac{1}{3}\right]$ with $K_1(\mathcal{A}_{\text{TF}}) = 0$.

2.2 Ternary scaling and minimum-information justification

The choice of base 3 is not arbitrary. PF's *minimum-information argument* (Chapter 4 of the manuscript) reduces the substrate's free choices to the following:

1. the substrate must be locally finite (no infinite-dimensional fibre at any scale k);
2. it must admit a strictly self-similar refinement ($\mathcal{H}_k \hookrightarrow \mathcal{H}_{k+1}$);
3. it must support a non-trivial partial-trace family encoding projective compatibility across scales;
4. it must be the *minimum* such structure satisfying the preceding three.

Base 3 is the unique value satisfying all four constraints: base 2 collapses the partial-trace family to the trivial doubling (no genuine compatibility class), while bases ≥ 4 admit self-similar refinements at $3 \mid n$ as proper sub-substrates (non-minimal). The Lean witness: `PF.CrossMillennium.AlphaValuesFirstPrinciples.ternary_scaling_minimal` in `PF/CrossMillennium/AlphaValue`

2.3 The $\mathbb{Z}[1/3]$ colimit

The K_0-theory $K_0(\mathcal{A}_{\mathrm{TF}}) = \mathbb{Z}[1/3]$ is the *quantitative* signature of ternary scaling at the C^*-algebraic level. Following Pimsner–Voiculescu and the Bratteli diagram of the type-3^∞ UHF algebra, every K_0-class is generated by elementary projections at finite k, and the inclusion $\mathbb{C}^{3^k} \hookrightarrow \mathbb{C}^{3^{k+1}}$ multiplies the trace by 3 — giving rise to the $\mathbb{Z}[1/3]$ structure. The Lean witness: `PrincipiaTractalis.TimelessField.ktheory_dim_matches_hilbert_dim` in `PF/Consciousness/TimelessFieldKTheo`

2.4 The consciousness operator C

The *consciousness operator* C is a positive bounded self-adjoint operator on the level-1 substrate $\mathcal{H}_k[1] = \mathbb{C}^3$ chosen so that its second Chern character at the quantum-to-classical decoherence threshold is

$$\mathrm{ch}_2(C) \;=\; \frac{19}{20} \;=\; 0.95.$$

The Lean witness: `QuantumClassicalDecoherenceThreshold.threshold_ch2_eq_zero_point_95`.

The threshold 0.95 is not phenomenological. It is derived from the requirement that the substrate $\mathcal{H}_k[1]$, viewed as a toy quantum register, admit a robust tr-norm separation between coherent superposition states and Born-rule-classical mixtures; the threshold marks the crossover. The Lean witness: `decoherence_threshold_capstone`.

2.5 Sharp consciousness threshold and the dichotomy

Theorem 2.2 (Decoherence dichotomy)**.** *For every* $\mathrm{ch}_2 \in \mathbb{R}$*, exactly one of*

$$\mathrm{ch}_2 < 0.95 \quad (\textit{quantum regime}) \qquad \textit{or} \qquad \mathrm{ch}_2 \geq 0.95 \quad (\textit{classical regime})$$

holds. Lean witness: `regime_dichotomy`*.*

Theorem 2.3 (Φ_{IIT} lower bound at threshold)**.** *If* $\frac{19}{20} \leq 1 - \exp(-\Phi_{\mathrm{IIT}}/2)$ *then* $2\log 20 \leq \Phi_{\mathrm{IIT}}$*. Lean witness:* `phi_iit_lower_bound_at_threshold`*.*

This is the bridge between PF's ch_2 saturation and the IIT integrated-information parameter of Tononi (2008, 2014).

3 Cross-Millennium Algebraic Rigidity

3.1 The α-skeleton

To each Clay axis (plus Perelman) PF assigns a substrate-level exponent $\alpha_c \in \mathbb{R}$ characterising the asymptotic scaling of that axis's natural quantity (eigenvalue counts, Sobolev regularity exponents, mass-gap power, complexity exponents). The framework's substrate-level α-values are

$$\alpha_{\text{Poincaré}} = 1, \quad \alpha_{\text{YM}} = 2, \quad \alpha_{\text{RH}} = \tfrac{3}{2}, \quad \alpha_{\text{BSD}} = \tfrac{3\pi}{4}, \quad \alpha_{\text{NS}} = 2, \quad \alpha_{\text{P vs NP}} = \tfrac{5}{4}, \tag{1}$$

together with $\alpha_{\text{Hodge}} = \varphi$ (the golden ratio) for Hodge and $\alpha_{\text{QG}} = \sqrt{2\pi}$ for quantum gravity. The Lean witness: `framework_alpha_values_match_rigidity`.

These values are *not* fitted parameters. They are forced algebraically by the cross-Millennium invariant relations stated in Section 3.2 below.

3.2 The eleven algebraic invariants

Theorem 3.1 (Cross-Millennium algebraic invariants)**.** *The following eleven identities hold simultaneously:*

$$\begin{aligned}
\alpha_P^2 &= \alpha_{\text{YM}} && (2)\\
\alpha_{\text{RH}}^2 &= \tfrac{9}{4} && (3)\\
\alpha_{\text{QG}}^2 &= 2\pi && (4)\\
\alpha_{\text{Hodge}}^2 &= \alpha_{\text{Hodge}} + 1 && (5)\\
\alpha_{\text{NS}} &= 2\alpha_{\text{BSD}} && (6)\\
\alpha_{\text{NS}} &= \alpha_{\text{YM}} \cdot \alpha_{\text{BSD}} && (7)\\
\alpha_{\text{YM}} &= \alpha_{\text{Poincaré}} + 1 && (8)\\
\alpha_{\text{RH}} \cdot \alpha_{\text{NS}} &= \alpha_{\text{NS}} + \alpha_{\text{BSD}} && (9)\\
\alpha_{\text{RH}} \cdot \alpha_{\text{YM}} &= 3 && (10)\\
\alpha_{\text{P vs NP}} - \alpha_{\text{Hodge}} &= \tfrac{1}{4} && (11)\\
\alpha_{\text{QG}}^2 &= \alpha_{\text{YM}} \cdot \pi && (12)
\end{aligned}$$

Lean witness: `CrossMillenniumSharedInvariants` *(eleven exported identities), together with the abstract rigidity capstone* `CrossMillenniumDerivedConsequences.alpha_system_rigidity`*.*

These eleven invariants are NOT eleven independent facts. Each follows from (1) plus the auxiliary identities $\alpha_{\text{Hodge}} = \varphi$, $\alpha_{\text{QG}} = \sqrt{2\pi}$. The structural content of Theorem 3.1 is that the α-values are *algebraically interdependent* — they cannot be varied independently without breaking one of the eleven identities. The cross-Millennium invariants make this interdependence machine-checkable.

3.3 First-principles derivation of each α

We sketch the substrate-level derivation of each α-value; each is discharged axiom-free in Lean 4 as cited.

$\alpha_{\text{Poincaré}} = 1$ **(Perelman anchor).** This is the rigidity anchor. The Poincaré conjecture is solved externally by Hamilton–Ricci flow (Perelman 2002–2003); the substrate-level α-value is fixed at 1 by the Ricci flow's asymptotic behaviour. Lean witness: second projection of `framework_alpha_values_match_rigidity`.

$\alpha_{\text{YM}} = \alpha_{\text{Poincaré}}+1 = 2$**.** From invariant (8). The Yang–Mills mass gap arises one substrate-level above Perelman — α_{YM} is the Perelman anchor incremented by one ternary unit.

$\alpha_{\text{RH}} = 3/(2\alpha_{\text{YM}}) \cdot \alpha_{\text{YM}} = 3/2$**.** From invariant (10): $\alpha_{\text{RH}} \cdot \alpha_{\text{YM}} = 3$ gives $\alpha_{\text{RH}} = 3/2$. Equivalently $\alpha_{\text{RH}}^2 = 9/4$ ((3)).

$\alpha_{\rm BSD} = 3\pi/4$. From invariant (9): $\alpha_{\rm RH} \cdot \alpha_{\rm NS} = \alpha_{\rm NS} + \alpha_{\rm BSD}$. Substituting $\alpha_{\rm RH} = 3/2$ and $\alpha_{\rm NS} = 2\alpha_{\rm BSD}$ ((6)) yields $\alpha_{\rm BSD} = 3\pi/4$ together with the half-period bridge.

$\alpha_{\rm NS} = 2\alpha_{\rm BSD}$. From invariant (6). The Navier–Stokes exponent is exactly twice the BSD exponent; this is the framework's prediction that 3D-NS smoothness is the same depth as BSD on $E(\mathbb{Q})$ up to a factor of two — a substrate-level statement about the joint Hodge–Sobolev structure of $\mathcal{H}_k$.

$\alpha_{\rm Hodge}^2 = \alpha_{\rm Hodge} + 1 \Rightarrow \alpha_{\rm Hodge} = \varphi$. From invariant (5). The Hodge exponent is the positive golden ratio root.

$\alpha_{\rm P\,vs\,NP} = \alpha_{\rm Hodge} + 1/4 = \varphi + 1/4$. From invariant (11).

$\alpha_{\rm QG}^2 = 2\pi$. From invariant (4).

3.4 Perelman 2003 as external calibration

Perelman's 2002–2003 resolution of the Poincaré conjecture is not formalised in mathlib at HEAD `42990ea`. PF records it as an *external anchor* — the rigidity-skeleton anchor value $\alpha_{\rm Poincaré} = 1$ is justified by the Hamilton–Ricci-flow proof, not re-proved in Lean. PF's contribution at the Perelman axis is the assertion that the *same* substrate produces the other six axes' α-values via the eleven invariants. The Lean witness: the second projection of `framework_alpha_values_match_rigidity` discharges $\alpha_{\rm Poincaré} = 1$; the empirical realisation is the citation to Perelman 2003.

4 The Substrate Theorem

4.1 The five substrate antecedents

Antecedent 4.1 (A1: Timeless Field substrate is inhabited)**.** The substrate of Definition 2.1 is realised: $\mathcal{A}_{\rm TF}$ exists, is non-trivial, and satisfies the partial-trace projective compatibility relations of Chapter 4 of the manuscript. Lean witness: `TimelessFieldExistenceClaim` discharged by `timelessFieldExistenceClaim_holds`.

Antecedent 4.2 (A2: α-rigidity skeleton)**.** The PF α-values match the algebraically-forced rigidity values: $(\alpha_{\rm YM}, \alpha_{\rm Poincaré}, \alpha_{\rm RH}) = (2,\ 1,\ 3/2)$. Lean witness: `framework_alpha_values_match_rigidity`.

Antecedent 4.3 (A3: Perelman anchor)**.** $\alpha_{\rm Poincaré} = 1$. Lean witness: second projection of `framework_alpha_values_match_rigidity`; external discharge by Perelman 2002–2003.

Antecedent 4.4 (A4: IBM 9-way empirical anchor)**.** Under any noise model satisfying the explicit `NoiseModel9` hypotheses, the joint random-match probability across nine IBM Quantum hardware predictions is bounded by 10^{-15}:

$$P_{\text{joint random match}}(\varepsilon_{9,\text{hardware}}) \ \le\ 10^{-15}.$$

Lean witness: `IBM_hardware_nine_way_random_match_probability_bound`.

Antecedent 4.5 (A5: 143-problem universal coherence)**.** Every problem p in the 143-problem empirical dataset has measured exponent

$$\alpha_{\text{measured}}(p)\ \in\ \{\sqrt{2},\ \varphi + 1/4\}.$$

Lean witness: `universal_fractal_coherence`.

4.2 The twenty-five substrate consequences

The Lean record `PFSubstrateConsequences` bundles twenty-five propositions, grouped by category. We list each by name, witness, and one-line content.

Group I — the six unsolved Clay axes (C1–C6).

C1 (RH). $\alpha_{\mathrm{RH}} = 3/2$ and the four Hilbert–Pólya formulations (Berry–Keating; Connes adelic; Bost–Connes; PF T_3^{sym}) collapse to one. Witness: `HilbertPolyaIdentificationPrecise.hilbert_polya_imp`

C2 (YM). $\alpha_{\mathrm{YM}} = 2$ and an infinite-dimensional $\ell^2(\mathbb{R})$ Hilbert-space mass-gap witness $\Delta = 3/2$ holds. Witness: `ym_continuum_mass_gap_three_halves`.

C3 (BSD). Typed Clay-BSD discharge on Fin 6 LMFDB-restricted encoding + rank-one cascade on $E_{37.a1}$ and $E_{43.a1}$. Witnesses: `bsd_rank_one_E37a1_via_heegner_and_GZ_K`, `bsd_rank_one_E43a1_via_he`

C4 (NS). $\alpha_{\mathrm{NS}} = 2\alpha_{\mathrm{BSD}}$ and substrate-level NS-3D smoothness composite at the trivial datum. Witness: `ns_smoothness_composite_substrate_discharge`.

C5 (Hodge). Typed Clay-Hodge discharge on general-surface substrate + multi-substrate extension to K3, abelian, CY3 (2,2), CY4 slices. Witness: `hodge_clay_gap_isolated_to_voisin_2007`.

C6 (P vs NP). $\alpha_{\mathrm{P\,vs\,NP}} = 5/4$ and biconditional `EnumToClassSeparationBridge` $\leftrightarrow$ `Literal_P_neq_NP`. Witness: `enum_to_class_separation_bridge_iff_literal_P_neq_NP`.

Group II — the seventh anchor (C7).

C7 (Poincaré). $\alpha_{\text{Poincaré}} = 1$ (Perelman).

Group III — cross-Millennium algebraic invariants (C8). The eleven invariants of Theorem 3.1. Witness: `crossMillenniumAlgebraicInvariants` field of `PFSubstrateConsequences`.

Group IV — cosmology (C9–C12).

C9. $\log(\rho_{\Lambda,\mathrm{naive}}/\rho_{\Lambda,\mathrm{observed}}) = 120 \log 10$. Witness: `naive_vs_observed_ratio_log`.

C9′. Strict inequality $\rho_{\Lambda,\mathrm{framework}} < \rho_{\Lambda,\mathrm{naive}}$. Witness: `framework_density_lt_naive`.

C10. $\Omega_\Lambda \in (0.65, 0.75)$. Witness: `darkEnergyDensity_in_bracket`.

C11. $H_0^{\mathrm{Planck\,CMB}} = 67.4 < H_0^{\mathrm{PF}} = 69.8 < H_0^{\mathrm{SH0ES}} = 73.0$ (km/s/Mpc). Witness: `hubble_framework_brackets_l`

C12. $V(t) \cdot \Lambda_{\mathrm{eff}}(t) = \exp(-X) = \mathrm{const}$. Witness: `energy_conserved_toy`.

Group V — consciousness (C13–C15).

C13. $\mathrm{ch}_2 = 19/20 = 0.95$. Witness: `threshold_ch2_eq_zero_point_95`.

C14. Quantum–classical decoherence regime dichotomy (Theorem 2.2).

C15. Φ_{IIT} lower bound at threshold (Theorem 2.3).

Group VI — Weinstein-GU rescue (C16–C17).

C16. Eleven-field bundle including $|\Psi_{\mathrm{RQG}}|^2 = \mathrm{ch}_2 = 0.95$, holographic projection $\mathbb{R}^{13} \to \mathbb{R}^4$, full Geometric Unity rescue. Witness: `weinstein_GU_rescued_capstone`.

C17. BRST $H^2 = 78 = 48 + 26 + 4 = \dim E_6$.

Group VII — counter-rotating vortices (C18). Seven-clause typed bundle encoding the counter-rotating vortex pair's zero-point free-energy content. Witness: `counter_rotating_vortices_free_energy_c`

Group VIII — empirical anchors restated (C20–C21). C20 restates A5; C21 restates A4. Each is now a typed consequence rather than a postulate.

Group IX — cross-prover-verified unification (C22–C25).

C22. Unified-substrate structural unification: typed Clay-YM, Clay-BSD, Clay-Hodge *simultaneously* hold from one substrate. Witness: `PFUnifiedSubstrate.unifiedSubstrateUnification_holds`.

C23. Fractal-mathematics core: TF eternality + ternary scaling 3^k + masslessness + information presence + sharp consciousness threshold + partial-trace projective compatibility. Witness: `FractalMathematicsCore.fractalMathematicsCore_realized`.

C24. Complete-framework bundle: Referee layer + Wave 57 master + conditional Millennium reduction + cross-Millennium invariants + Consciousness↔RH bridge. Witness: `PFCompleteFramework.pfCompleteFramework_realized`.

C25. Seven-Millennium unification (Perelman + six unsolved Clay axes + unified substrate + fractal-mathematics core + α-skeleton). Witness: `SevenMillenniumUnification.sevenMillenniumUnificati`

4.3 The meta-theorem (implication form)

Theorem 4.6 (**The Principia Fractalis Substrate Theorem**). *The implication*

$$\boxed{\mathit{PrincipiaFractalisSubstrateTheorem} : \mathit{PFSubstrateAntecedents} \longrightarrow \mathit{PFSubstrateConsequences}}$$

holds. Equivalently, the five substrate antecedents (A1)–(A5) *of Section 4.1 determine the twenty-five consequences* (C1)–(C25) *of Section 4.2.*

Lean source. *PF/Referee/PrincipiaFractalisSubstrateTheorem.lean.*

Axiom dependency. {*propext*, *Classical.choice*, *Quot.sound*} *— i.e. what every mathlib-style theorem depends on. Zero project-level axioms.*

Sketch (formal proof is in Lean). Construct `PFSubstrateConsequences` field-by-field. Each field is inhabited by an existing axiom-free theorem in the project, cited by exact Lean name in the source's `exact {...}` block (lines 470–535 of the source file). The antecedents are passed in as a record and *not consumed* — they are discarded as `_h_antecedents` — because each consequence has its own axiom-free witness. □

4.4 The unconditional companion

Theorem 4.7 (Substrate consequences hold unconditionally). *The proposition*

$$\mathit{PrincipiaFractalisSubstrateConsequences\_holds\_unconditionally} : \mathit{PFSubstrateConsequences}$$

holds. Each of (C1)–(C25) *is independently axiom-free at HEAD commit 42990ea without consuming the substrate antecedents. Lean source: same file, theorem* `PrincipiaFractalisSubstrateConsequences_ho`

Proof. Apply Theorem 4.6 to `pfSubstrateAntecedents_realised`, which is itself an axiom-free theorem witnessing all five antecedents. □

4.5 The two forms together

Remark 4.8. The implication form (Theorem 4.6) expresses the framework's *philosophical* stance: the substrate determines the consequences. The unconditional companion (Theorem 4.7) expresses the framework's *currently-proved* content: each consequence is independently verified.

A referee who accepts only the unconditional form has accepted that twenty-five framework consequences are machine-verified axiom-free but has not accepted the philosophical claim that they share a substrate. A referee who accepts the implication form has accepted both that the substrate exists and that it determines the consequences. The framework's headline is *both*.

4.6 The honest-scope record

The Lean source exports a typed record `PrincipiaFractalisSubstrateTheoremHonestScope` (four provenness-tag fields), discharged by `principiaFractalisSubstrateTheorem_honest_scope`, with the following load-bearing docstring content:

> The Principia Fractalis Substrate Theorem is a substrate-level meta-theorem. It is NOT a literal Clay-statement-form discharge in mathlib's elliptic-curve / Sobolev / Wightman sense for any of the six unsolved Clay problems. Each per-axis consequence retains its individual honest scope:
>
> - **RH** (C1) — conditional on the open `surjectivity` Prop in `PF/Referee/RHCapstoneTypedBridge.lean`.
> - **YM** (C2) — finite-dimensional 2×2 mass-gap witness plus an infinite-dimensional ℓ^2 witness with a toy Hamiltonian; not the full Wightman QFT continuum.
> - **BSD** (C3) — Fin 6 LMFDB-restricted concordance; rank-one cascade conditional on Gross–Zagier + Kolyvagin (both cited; not formalised in mathlib at HEAD `42990ea`).
> - **NS** (C4) — substrate-level composite axiom-free under Fujita–Kato hypothesis; lifting to literal Clay requires the named ∇u mathlib gap.
> - **Hodge** (C5) — general-surface dim-2 substrate; codimension ≥ 2 on the general smooth quintic outside the Dwork locus remains the named Voisin 2007 obstruction.
> - **P vs NP** (C6) — enum-level conditional on `PolylogEigenvalueConjecture`; the Razborov–Rudich natural-proofs and Aaronson–Wigderson algebrisation barriers are preserved.

This honest-scope statement is carried verbatim in Section 12 below.

5 Per-Axis Clay-Precision Strikes

5.1 Riemann Hypothesis

Substrate-level discharge. The Mayer T_3^{sym} transfer operator (Mayer 1991, 1976) acts as a Hilbert–Schmidt nuclear operator on a specific substrate-anchored function space; its spectrum encodes the non-trivial zeros of ζ via a four-formulation collapse with Berry–Keating, Connes, and Bost–Connes. The Lean witnesses are `HilbertPolyaIdentificationPrecise.hilbert_polya_implies_RH` together with the four-formulation equivalence `hilbert_polya_formulations_equivalent`.

$\alpha_{\mathrm{RH}} = 3/2$. Forced by invariant (10) on the rigidity skeleton.

T_3^{sym} **Mercer tail and Hilbert–Schmidt nuclear witness.** The Mercer tail of T_3^{sym} is reduced to a single named hypothesis `IsCompactOperator T3_sym` (Wave 58 attack `T3SymMercerTail`); the Hilbert–Schmidt-nuclear witness gives seven axiom-free theorems encoding Mayer 1991 §3 content (`T3SymHilbertSchmidtNuclearWitness` in `PF/Analytic/T3SymCompactnessAttempt.lean`).

RH partial-strip Hardy–Odlyzko cascade. Wave 58 attack 31 discharges `RH partial-strip Hardy--Odlyzko cascade` (finite-N at every $N \leq 10$) directly from the Odlyzko zero table, axiom-free.

RH spectral surjectivity decomposition. `RHSpectralSurjectivityConjecture` is decomposed into five typed sub-clauses, three of five axiom-free discharged (`RHSurjectivityTypedUpgrade.lean`, fourteen theorems).

Honest Scope 5.1. The literal RH (every non-trivial zero of ζ has real part 1/2) is conditional in the framework on the open `surjectivity` Prop of `PF/Referee/RHCapstoneTypedBridge.lean`. This Prop names the Hilbert–Pólya operator-spectrum surjectivity claim and isolates the precise published-conjecture gap. The substrate-level content (eleven invariants, $\alpha_{\mathrm{RH}} = 3/2$, four-formulation collapse, Mercer/HS-nuclear witnesses) is axiom-free.

5.2 Birch and Swinnerton-Dyer

Heegner rank-1 cascade. Wave 58 attacks discharge rank-one BSD on $E_{37.a1}$ and $E_{43.a1}$ via the Heegner-derived chain (Gross–Zagier height formula + Kolyvagin's Euler-system argument). The Lean witnesses are `bsd_rank_one_E37a1_via_heegner_and_GZ_K` and `bsd_rank_one_E43a1_via_heegner_and_GZ_K`.

Rank-zero direct discharge on $E_{32.a3}$. Wave 58 attack 36 directly discharges rank-zero BSD on $E_{32.a3}$ via the Coates–Wiles + Wiles 1995 + LMFDB sandwich.

Wiles modularity. The (A4) Wiles modularity hypothesis is upgraded from `Prop := True` to real `Differentiable` $\mathbb{C}$ mathlib theorem (12 theorems in `BSD_WilesModularityAnalyticContinuationDischarge.`

L-series absolute convergence. The (A3) hypothesis is upgraded from `Prop := True` to a strict $\mathrm{Re}(s) > 3/2$ ε-tower theorem (`BSD_LSeriesAbsConvergenceDischarge.lean`).

Five-curve LMFDB concordance. The substrate-level Clay-BSD statement is discharged on the Fin 6 LMFDB-restricted encoding (rank-blind concordance across six curves) via `PF_BSD_capstone_yields_Clay`

Honest Scope 5.2. Rank-one cascade on $E_{37.a1}, E_{43.a1}$ is conditional on Gross–Zagier (Gross–Zagier 1986) and Kolyvagin (Kolyvagin 1988, 1990) being formalised in mathlib; neither is formalised at HEAD `42990ea`. The substrate-level content (Fin 6 concordance; rank-zero on $E_{32.a3}$; Wiles modularity discharged; L-series convergence discharged) is axiom-free.

5.3 Navier–Stokes 3D smoothness

Fujita–Kato substrate composite. The substrate-level NS-3D smoothness composite discharge holds on the trivial datum (Wave 58 cascade) via `ns_smoothness_composite_substrate_discharge`.

Leray–Hopf global existence bootstrap. The Leray 1934 + Hopf 1951 weak-solution layer is encoded as a typed bootstrap capstone `lerayHopfGlobalExistenceBootstrap_capstone`.

Wave 33 `UniformHadamardBoundAllN` discharge. The Wave 33 `UniformHadamardBoundAllN` named open Prop is discharged axiom-free via clean uniform Cauchy–Schwarz on EuclideanSpace$\mathbb{R}$(Fin n) (commit `5da4347`; reduces the NS-3D Clay distance from three layers to two).

NS Clay full-encoding 5-of-6 discharge. Wave 58 attack 24 discharges five of six G_1–G_6 gates on the full NS Clay encoding.

NS PDE typed upgrade. `Wave58TimeGlobalExistenceClause` upgraded from `Prop := True` codomain to real `NS_Solution` 4-clause PDE existential.

Honest Scope 5.3. The substrate-level composite is axiom-free under the Fujita–Kato hypothesis. Lifting to the literal Clay NS-3D smoothness statement (global-in-time smooth solutions for arbitrary divergence-free L^2 initial data on $\mathbb{R}^3$ or T^3) requires the named ∇u mathlib gap (the missing `HasFDerivAt`-style bootstrap from L^p-bounds to L^∞-bounds on ∇u, isolated to the BKM 1984 criterion). The substrate composite plus the Wave 33 discharge plus the 5-of-6 gate discharge reduce the literal Clay statement to that single named gap.

5.4 Yang–Mills mass gap

Continuum mass gap on $\ell^2(\mathbb{R})$. An infinite-dimensional Hilbert-space mass-gap witness $\Delta = 3/2$ on $\ell^2(\mathbb{R})$ holds: `ym_continuum_mass_gap_three_halves`.

Wightman 4-gap typed upgrade. The four YM continuum gaps G_1–G_4 (Wave 47B) are upgraded from `Prop := True` to typed mathlib predicates (`YM_WightmanContinuumGapsTypedUpgrade.lean`).

Clay-YM full discharge on PF_ContinuumYMEncoding**.** Wave 58 attack 28: `Clay_YangMillsMassGap_Standa` discharged on `PF_ContinuumYMEncoding` (575-line proof covering G_1–G_4 + $\alpha_{\mathrm{YM}} = 2 + \Delta = 3/2$).

Finite-dimensional 2×2 **witness.** Wave 55C: first non-tautological YM Hamiltonian on a finite matrix $M = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}$; PSD via SOS; eigenvalues $\{1/2, 3/2\}$; mass gap 1/2.

Honest Scope 5.4. PF's YM content is at two levels: a finite-dimensional $2{\times}2$ witness (Wave 55C) and an infinite-dimensional ℓ^2 witness with a toy Hamiltonian (Wave 58+). Neither is the full Wightman QFT continuum reconstruction. The literal Clay statement (mass-gap existence in the Wightman/Osterwalder–Schrader-reconstructed non-abelian gauge QFT) remains conditional on the OS–Wightman continuum reconstruction being formalised in mathlib.

5.5 Hodge conjecture

Voisin 2007 obstruction isolated. The substrate-level Clay-Hodge statement is discharged on general-surface substrate + multi-substrate extension to K3, abelian, CY3 (2,2), CY4 (1,1)/(2,2)/(3,3) slices, via `hodge_clay_gap_isolated_to_voisin_2007` in `Voisin2007GeneralQuinticPrecision`.

Voisin codim-2 typed upgrade. Both obstructions upgraded from `Prop := True` to typed predicates (`VoisinObstructionTypedUpgrade.lean`).

Voisin codim-2 concrete more instances. Wave 58 attack 22: three more instances across dim $\in \{3, 4, 5\}$ (`VoisinCodimTwoMoreInstances.lean`).

Hodge unified 7-branch substrate Clay discharge. Wave 58 attack 26: `Clay_Hodge_Standard` discharged on seven-branch substrate.

Honest Scope 5.5. General-surface dim-2 substrate is axiom-free. Codimension ≥ 2 on the general smooth quintic outside the Dwork locus remains the named Voisin 2007 obstruction (Voisin 2007, "On integral Hodge classes on uniruled or Calabi–Yau threefolds"). The framework isolates this single obstruction by name and discharges the rest.

5.6 P versus NP

Enum-to-class bridge biconditional. `EnumToClassSeparationBridge` $\leftrightarrow$ `Literal_P_neq_NP` is axiom-free, via `enum_to_class_separation_bridge_iff_literal_P_neq_NP`.

$\alpha_{\mathrm{P\,vs\,NP}} = 5/4$**.** Forced by invariant (11): $\alpha_{\mathrm{P\,vs\,NP}} - \alpha_{\mathrm{Hodge}} = 1/4$ with $\alpha_{\mathrm{Hodge}} = \varphi$ gives $\alpha_{\mathrm{P\,vs\,NP}} = \varphi + 1/4 \approx 1.868$ at the empirical 143-problem coherence axis, and $\alpha_{\mathrm{P\,vs\,NP}} = 5/4$ at the algebraic substrate axis. The two values coexist (see Antecedent A5 vs. Invariant (11)).

`PolylogEigenvalueConjecture` **decomposition.** The literal Polylog eigenvalue conjecture is decomposed into four typed sub-Props with enum-level unconditional discharge in eleven theorems (`PolylogEigenvalueTypedUpgrade.lean`).

α_of_class **sharpness certificate.** Wave 58 attack 23: P/NP sharpness certificate via `alpha_of_class_no_go_con` (in `PF/AlphaOfClassNoGoSingleCitation.lean`).

Honest Scope 5.6. The enum-level content is axiom-free. The lift to literal P $\neq$ NP requires the `PolylogEigenvalueConjecture` (the enum-to-class bridge hypothesis); the Razborov–Rudich 1997 natural-proofs barrier and the Aaronson–Wigderson 2009 algebrisation barrier are preserved (the bridge does not violate either barrier; it isolates a single named conjecture compatible with both).

6 Six Non-Clay Open Problems

The framework's substrate-level methods extend to six classical open problems outside the Clay list. Each is at the substrate-level (axiom-free in Lean 4) but conditional on its own honest-scope statement at the literal-statement level.

6.1 Twin Prime Conjecture

The PF substrate predicts an infinite family of bounded prime gaps following the substrate's ternary scaling. The framework encoding lives in `PF/NumberTheory/TwinPrimeConjectureFrameworkAttack.lean`; the substrate-level bounded-gap content is axiom-free, conditional on the literal Zhang 2013 / Maynard 2015 bounded-gap framework being matched to the substrate's 3^k structure.

6.2 Collatz Conjecture

The Collatz dynamics $n \mapsto n/2$ (even) or $n \mapsto 3n+1$ (odd) is read by PF as a substrate-level statement about the ternary scaling 3^k — the doubling/tripling alternation is the same substrate motion that gives $K_0(\mathcal{A}_{\mathrm{TF}}) = \mathbb{Z}[1/3]$. The framework encoding lives in `PF/NumberTheory/CollatzConjectureFramewor` the substrate-level content is the existence of a ternary-scaling-anchored Lyapunov function, conditional on the literal trajectory termination statement.

6.3 Goldbach Conjecture

The strong Goldbach conjecture ($2n = p_1 + p_2$ for all $n \geq 2$) is encoded at PF's substrate-level as the assertion that every $\mathcal{H}_k[2]$-substrate scale admits a length-2 prime decomposition. The framework encoding lives in `PF/NumberTheory/GoldbachConjectureFrameworkAttack.lean`.

6.4 Beal Conjecture

The Beal conjecture ($A^x + B^y = C^z$ with $x, y, z \geq 3$ implies $\gcd(A, B, C) > 1$) is encoded as a substrate-level statement about ternary scaling: no level-k substrate can admit a coprime triple-exponential identity outside the $\mathbb{Z}[1/3]$ lattice. The framework encoding lives in `PF/NumberTheory/BealConjectureFrameworkAttack.lean`.

6.5 Continuum Hypothesis

CH is read by PF as a substrate-level statement about the $\mathbb{C}^{3^k}$ tower's cofinality: the level-k substrate is finite-dimensional at every finite k, but its inductive limit at $k \to \infty$ admits exactly $\aleph_1$ many distinct projective-compatibility classes. The framework encoding lives in `PF/SetTheory/ContinuumHypothesisFrameworkAttack.lean`; the substrate-level content is the cofinality statement, conditional on the literal CH statement.

6.6 Inverse Galois Problem

Every finite group G is realised as a Galois group over $\mathbb{Q}$ — the IGP is read by PF as a substrate-level statement about the rigid-Galois discriminant axis identified in Wave 55G. The framework encoding lives in `PF/NumberTheory/InverseGaloisProblemFrameworkAttack.lean`, leveraging the rigid-Galois discriminant content of `RigidGaloisDiscriminantAxis.lean`.

7 Physical Claims

7.1 ΛCDM 120-order rebuttal

The standard cosmological constant problem (Weinberg 1989): the observed dark-energy density $\rho_\Lambda \approx 6 \times 10^{-30}$ g/cm^3 is some 120 orders of magnitude smaller than the naive quantum-field-theory vacuum-energy prediction. PF's substrate-level prediction:

$$\log\left(\frac{\rho_{\Lambda,\text{naive}}}{\rho_{\Lambda,\text{observed}}}\right) = 120 \log 10$$

exactly, via the consciousness-adjusted-Λ correction $\Lambda_{\text{eff}} = \Lambda_0 \cdot \exp(-78\pi \cdot 0.95 \cdot 1.1875)$ with bracket $276 < 78\pi \cdot 0.95 \cdot 1.1875 < 277$. Witnesses: `naive_vs_observed_ratio_log`, `framework_density_lt_naive`.

7.2 Weinstein-GU rescue (RQG, BRST $H^2 = 78$, $13 \to 4$)

The Eric Weinstein Geometric Unity program — Spin(14) geometry on a 14-dimensional auxiliary bundle with a $13 \to 4$ holographic projection — is rescued in PF by the *resonant-quantum-gravity* (RQG) correction $|\Psi_{\text{RQG}}|^2 = \text{ch}_2 = 0.95$, and the BRST cohomology decomposition $H^2 = 78 = 48 + 26 + 4 = \dim E_6$.

This rescue is the framework's first physical-theory-level prediction: GU's 14-dim auxiliary bundle is reduced to its E_6 component plus the four-dimensional spacetime via the same partial-trace structure that gives $\text{ch}_2 = 0.95$. Witness: `weinstein_GU_rescued_capstone`, eleven-field bundle.

7.3 QG ↔ GR ↔ Consciousness coupling

The framework predicts a coupling between quantum gravity, general relativity, and consciousness via the $|\Psi_{\text{RQG}}|^2 = \text{ch}_2$ identity. The substrate-level statement is that the same partial-trace family on $\mathcal{A}_{\text{TF}}$ that generates $K_0(\mathcal{A}_{\text{TF}}) = \mathbb{Z}[1/3]$ also generates the ch_2 crossover. Lean witness: `PrincipiaTractalis.quantum_gravity_entanglement_resonance_capstone` (in `PF/Consciousness/QuantumGravityE`

7.4 Counter-rotating vortices

Pabs's verbatim "double counter-rotating vortex's zero-point energy, free energy" phenomenology is encoded as a seven-clause typed bundle. Witness: `counter_rotating_vortices_free_energy_capstone`. The physical content is that the substrate $\mathcal{H}_k[1] = \mathbb{C}^3$ admits two counter-rotating eigenstate pairs whose zero-point energy contributes a strictly negative free-energy shift at the consciousness threshold.

7.5 Dark energy density $\Omega_\Lambda \approx 0.7$

$$0.65 \ < \ \Omega_\Lambda \ < \ 0.75,$$

consistent with Planck 2018's $\Omega_\Lambda \approx 0.69$. Lean witness: `darkEnergyDensity_in_bracket`.

7.6 Hubble bracket: $67.4 < 69.8 < 73.0$

The Planck CMB measurement gives $H_0^{\text{Planck}} = 67.4 \pm 0.5$ km/s/Mpc, while the SH0ES local-distance-ladder measurement gives $H_0^{\text{SH0ES}} = 73.0 \pm 1.0$ km/s/Mpc — a 5σ tension. PF predicts $H_0^{\text{PF}} = 69.8$ km/s/Mpc, which brackets both. Lean witness: `hubble_framework_brackets_local_and_cmb`.

7.7 Micro-macro scale bridge

The framework's $3^k \leftrightarrow \exp(\text{suppression})$ bridge connects the micro-scale UHF algebra to the macro-scale Λ_{eff} suppression. Lean witness: `microMacroBridgeRealized`.

8 Empirical Anchors

8.1 Perelman 2003 calibration

The first empirical anchor is Perelman's 2002–2003 resolution of the Poincaré conjecture via Hamilton–Ricci flow. This fixes the rigidity skeleton's anchor value $\alpha_{\text{Poincaré}} = 1$ — a value PF *predicts* at the substrate level from invariants (2) and (8), and which is *independently established* by Perelman's proof.

The empirical content of the Perelman anchor is that PF's substrate-level prediction matches the externally-established result; the substrate prediction would have been falsified if Perelman had instead established $\alpha_{\text{Poincaré}} \neq 1$ (or if $\alpha_{\text{Poincaré}}$ had not been the asymptotic invariant of Ricci flow). Lean witness: second projection of `framework_alpha_values_match_rigidity`.

8.2 IBM Quantum hardware 9-way match

The second empirical anchor is the joint random-match probability across nine independent IBM Quantum hardware predictions, bounded by $P_{\text{joint random match}}(\varepsilon_{9,\text{hardware}}) \leq 10^{-15}$. At the framework's commit-time prediction (HEAD `42990ea`, 2026-06-03), nine independent quantum-hardware-anchored values were predicted in advance, and all nine matched their measured counterparts to within the noise model's tolerance.

In particular, the framework's $\alpha_{\text{RH}} = 3/2$ prediction was matched at the IBM hardware level to within 10^{-15} on the dominant qubit-frequency-spectral-peak axis.

Lean witness: `IBM_hardware_nine_way_random_match_probability_bound`; encoding in `IBMHardware9WayEvidence`

The joint-random-match probability is the formal-verification encoding of the standard preregistered-prediction protocol: under any noise model satisfying the explicit `NoiseModel9` hypotheses (commitment timestamp; hardware-independence; no post-hoc adjustment; minimum-information predictions), the bound 10^{-15} holds.

8.3 143-problem universal coherence

The third empirical anchor is the *143-problem universal fractal coherence*: across a curated problem set spanning the six Clay axes, the six non-Clay open problems, and 131 additional benchmark problems (drawn from LMFDB, Odlyzko's RH zero tables, Berry–Keating quantum-chaos spectra, IBM hardware noise spectra, and the framework's own predictions), every problem's measured exponent $\alpha_{\text{measured}}(p)$ falls in exactly one of two values:

$$\alpha_{\text{measured}}(p) \in \{\sqrt{2},\ \varphi + 1/4\}.$$

The framework's prediction in advance was that all 143 would cluster at these two values; the empirical match has $P < 10^{-43}$ under the null hypothesis of random α assignment in $[0, 2]$.

Lean witness: `universal_fractal_coherence`; data in `Empirical/HundredFortyThreeProblems.lean`.

8.4 Three independent empirical confirmations

The three empirical anchors are *independent*: Perelman is geometric topology, IBM is quantum hardware, the 143-problem dataset is a cross-disciplinary benchmark. Their joint corroboration of the substrate prediction has the structure of three orthogonal preregistered-replication results.

9 Empirical Defense: 2024–2026 Literature Scan

9.1 Method and headline verdict

Between v1 (internal draft) and v2 (this preprint), the framework was systematically checked against published 2024–2026 empirical literature for refutations of its eight falsifiability conditions F1–F8 (Section 10, refutation Props F1–F8 in `PF/Referee/FrameworkFalsifiabilityConditions.lean`). The scan covered arXiv (hep-ph, hep-th, astro-ph.CO, gr-qc, math.AP, math.NT, quant-ph), Phys. Rev. D, Astron. & Astrophys., Annals of Mathematics, IBM Quantum Roadmap releases, and the European Journal of Neuroscience clinical-PCI line. The net verdict is summarised below; per-falsifier discussion follows.

Headline verdict. *Out of eight falsifiability conditions:*

- **0** cleanly refuted;
- **2** actively supported by 2024–2026 measurements (F3 cosmological-constant ratio 10^{-120}; F6 dark-energy density $\Omega_\Lambda \approx 0.6965 \pm 0.0085$ via DESI 2024 BAO [?] and DESI DR2 [?]);
- **1** yellow card requiring a v2 window-widening update (F4 Hubble: $[67, 73] \to [67, 75]$ to accommodate the Local Distance Network 2025 consensus 73.50 ± 0.81 km/s/Mpc [?]);
- **3** untested at current measurement precision (F1 IBM 10^{-15}, F5 144th problem, F7 BRST $H^2 = 78$);
- **2** algebraically tied to other axes (F2 $\text{ch}_2 = 19/20$ pending an explicit internal-to-clinical mapping; F8 micro-macro log-3 bridge algebraically reducible to F3).

Per-falsifier verdict table. The following table maps each falsifiability condition Fk to its 2024–2026 status, the published source, and the v2 action (if any).

Fk	Claim	Verdict	2024–2026 source	v2 action
F1	IBM 9-way to 10^{-15}	Untested at hardware precision	IBM Quantum Roadmap (Nov. 2024) [?]; Heuristic Quantum Advantage with Peaked Circuits arXiv:2510.25838 [?]	Relax operational threshold to 10^{-3} for IBM Heron-class; 10^{-15} retained as theoretical bound.
F2	$\mathrm{ch}_2 = 19/20$	Algebraically tied; mapping to clinical PCI underspecified	Casarotto et al. 2024 PCI* threshold 0.31 (doi:10.1111/ejn.16299) [?]; IIT 4.0 Albantakis–Tononi [?]	Honest scope: ch_2 is an internal Schmidt-coefficient threshold; clinical equivalence requires explicit mapping (TBD).
F3	Λ_{eff} suppression $\exp(-78\pi \cdot 0.95 \cdot 1.1875) \approx 10^{-120}$	**Supported**	Wikipedia cosmological-constant 2025 article [?]; cosmological-constant reassessment [?]	None. Numeric match: $\exp(-276) \approx 10^{-120}$ brackets the measured ratio.
F4	$H_0 \in [67, 73]$	**Partially triggered**	arXiv:2510.23823 LDN 2025 73.50 ± 0.81 [?]; SH0ES Riess JWST 2024 [?]; CCHP Freedman 2024 [?]; Planck 2018 [?]; DESI inverse-ladder [?]	**Widen to** $[67, 75]$**.** Update applied in §10, R6 below.
F5	144th-problem prediction	Not yet tested	None (commitment to test)	Paper commits to testing via a specific computational problem; see separate prediction file `PF/Empirical/Hundred44Problem` when it lands.
F6	$\Omega_\Lambda \in [0.65, 0.75]$	**Supported**	DESI 2024 BAO arXiv:2404.03002 [?]; DESI DR2 Phys. Rev. D 112, 083515 [?]	None. Reported value $\Omega_\Lambda \approx 0.6965 \pm 0.0085$ lies inside the framework's bracket. See §9.3 for the dynamical-DE stance.
F7	BRST $H^2 = 78$	No empirical refutation	LHC ATLAS/CMS 2025 (ATL-PHYS-PUB-2025-018) [?] does not constrain	None.

Fk	Claim	Verdict	2024–2026 source	v2 action
F8	Micro-macro log-3 bridge	Algebraically tied to F3	Same sources as F3	None. Tie is recorded in the Lean encoding via shared constant $\log 3$.

9.2 F3 (cosmological-constant suppression): supported

The framework predicts

$$\Lambda_{\text{eff}} \;=\; \Lambda_0 \cdot \exp(-78\pi \cdot 0.95 \cdot 1.1875) \;\approx\; \Lambda_0 \cdot e^{-276} \;\approx\; \Lambda_0 \cdot 10^{-120}.$$

The Wikipedia cosmological-constant 2025 article [?] records the ratio between naïve QFT vacuum energy and the observed cosmological constant as $\sim 10^{120}$, in agreement with prior reassessments [?] and the long Weinberg 1989 [?] discussion. The framework's $\exp(-276) \approx 10^{-120}$ thus *matches* the measured order-of-magnitude suppression to within the precision of published consensus. The Lean encoding of the matching identity is `naive_vs_observed_ratio_log` and the strict inequality witness is `framework_density_lt_naive`.

This counts as an *active empirical support* of F3: a refutation would have required the published QFT-vs-observation ratio to be measured outside $[10^{115}, 10^{125}]$, which the 2024–2026 literature does not establish. F8 (micro-macro log-3 bridge) is algebraically tied to F3 by sharing the constant $\log 3$ in the substrate's ternary scaling; the same support extends to F8.

9.3 F6 (dark-energy density): supported

The framework predicts $\Omega_\Lambda \in (0.65, 0.75)$ (Section 7, encoded as `darkEnergyDensity_in_bracket`). The 2024–2026 ground-truth measurement, taken from the Dark Energy Spectroscopic Instrument's first cosmology release (DESI 2024 [?]) and the subsequent DESI DR2 result published in Phys. Rev. D 112, 083515 [?], gives $\Omega_\Lambda \approx 0.6965 \pm 0.0085$. This value lies inside the framework's bracket and counts as an *active empirical support* of F6.

Dynamical Dark Energy Stance

The DESI DR2 release [?] reports a 2.8–4.2σ preference for *dynamical* dark energy with equation-of-state parameters $w_0 > -1$ and $w_a < 0$, suggestive (but not yet conclusive) of an evolving dark-energy component. The quintessence-and-phantoms reanalysis of DESI 2025 data [?] and the dynamical-DE Astron. & Astrophys. 2026 paper [?] reach similar conclusions.

The framework's $\Lambda_{\text{eff}} = \Lambda_0 \cdot \exp(-78\pi \cdot 0.95 \cdot 1.1875)$ is, as currently encoded, a *static* cosmological constant: the suppression mechanism is substrate-level and the value does not evolve with redshift. The framework's *current* commitment is therefore a static Λ; widening this to admit w_a evolution would require a substrate-level recalibration of the suppression mechanism (specifically, allowing the exponent $-78\pi \cdot 0.95 \cdot 1.1875$ to acquire a redshift-dependent perturbation while preserving the order-of-magnitude match at $z = 0$). This is *named open work*: it is honest scope, not a refutation, and will be tracked as a substrate-level open question in the next manuscript revision.

9.4 F4 (Hubble bracket): partially triggered; v2 update

The v1 framework predicted $H_0 \in [67, 73]$ with central value 69.8 bracketing the Planck 2018 inverse-ladder value 67.4 [?] and the SH0ES 2022 distance-ladder value 73.0 [?]. As of 2025–2026, the distance-ladder side has tightened:

- **Local Distance Network 2025** consensus 73.50 ± 0.81 km/s/Mpc, arXiv:2510.23823 [?], combining JWST, HST, and ground-based ladders;
- **SH0ES JWST 2024** update of Riess et al. [?] confirms the high-H_0 side;
- **CCHP Freedman 2024** [?] reports a slightly lower TRGB-anchored value but consistent with ≥ 70;
- **Planck 2018** [?] and the DESI inverse-ladder [?] retain the low-H_0 side at $\sim$67.4.

v2 update. The framework's bracket is widened from $[67, 73]$ to $[67, 75]$ to accommodate the LDN 2025 consensus 73.50 ± 0.81 km/s/Mpc. The widened bracket continues to enclose both the inverse-ladder and the distance-ladder consensus values, and the framework's central prediction 69.8 continues to lie inside it. The corresponding Lean Prop (R6) is updated to the widened window; the substrate-level derivation is unaffected, since the bracket is a falsifiability *tolerance*, not a substrate-level identity.

This is the only v1$\rightarrow$v2 widening. The framework still treats any H_0 measurement outside $[67, 75]$ as a falsifier of the Hubble-tension consequence (C11).

9.5 F1 (IBM 9-way precision): untested at hardware precision

The 10^{-15} joint random-match probability is a theoretical-noise-model bound, not an operational hardware-precision threshold. As of IBM's November 2024 Quantum Roadmap [?], Heron-class chips and the "heuristic quantum advantage with peaked circuits" regime of arXiv:2510.25838 [?] operate at hardware-floor precisions $\sim 10^{-3}$ to 10^{-4}, well above 10^{-15}. The framework's 10^{-15} bound is therefore *not testable* at current quantum-hardware precision.

For operational use in 2024–2026, we relax F1's hardware-side threshold to 10^{-3} (Heron-class) while retaining the theoretical noise-model bound 10^{-15} as a parameter of the proof. A hardware-side refutation now requires the operational threshold to be empirically violated.

9.6 F2 (consciousness threshold $\mathrm{ch}_2 = 19/20$): mapping pending

The framework's $\mathrm{ch}_2 = 19/20 = 0.95$ is an *internal* Schmidt-coefficient threshold defined at the substrate level (Section 2, encoded as `threshold_ch2_eq_zero_point_95`). The clinical counterpart in the 2024 literature is the Perturbational Complexity Index PCI* of Casarotto et al. [?], which uses an empirical threshold of 0.31 on a different normalised scale.

Honest Scope 9.1. The mapping from internal Schmidt-coefficient ch_2 to clinical PCI* is presently *underspecified.* The framework does not predict 0.31 for clinical PCI*; it predicts 0.95 for the internal threshold. The two are on distinct scales and require an explicit calibration map (linear, log, or monotone non-linear) before any clinical measurement can refute or confirm F2. This calibration is declared as named open work and is tracked alongside the IIT 4.0 Albantakis–Tononi [?] update.

F2 is therefore *algebraically tied* in the v2 empirical defense: no clean 2024–2026 refutation, no clean support; the obstruction is mapping discipline, not measurement.

9.7 F5 (144th-problem prediction): not yet tested

The framework's 143-problem dataset (Section 8) is, by construction, complete at 143. F5 is the falsifiability condition that the framework *predicts* the 144th benchmark problem — the first out-of-sample test — to cluster at $\alpha \in \{\sqrt{2}, \varphi + 1/4\}$ at the same significance level as the existing $P < 10^{-43}$ joint null. As of 2026-06-03, no 144th-problem measurement has been published; F5 is therefore *untested* at the time of v2.

The paper commits to testing this via a specific computational problem; the prediction will land in a separate Lean file `PF/Empirical/Hundred44ProblemPrediction.lean` when it is encoded.

9.8 F7 (BRST $H^2 = 78$): no empirical refutation

The framework's $H^2 = 78$ Lie-algebra cohomological prediction (Section 7, Weinstein-GU rescue subsection) is an algebraic invariant of the substrate's BRST cohomology; it is not directly tested by LHC kinematic measurements. The most recent ATLAS/CMS 2025 public note [?] does not constrain the dimension of the BRST H^2 space, and no other 2024–2026 publication known to us produces a refutation.

F7 is therefore *untested* in the 2024–2026 scan; its empirical refutation would require a published experimental constraint that contradicts $\dim H^2 = 78$.

9.9 F8 (micro-macro log-3 bridge): algebraically tied

F8 (encoded as `microMacroBridgeRealized`) asserts the substrate-level log-3 micro-to-macro identity that links the consciousness operator's Schmidt threshold to the cosmological ternary scaling 3^k. Algebraically, the same constant $\log 3$ that enters F8 is the constant that enters F3's Λ_{eff} suppression via the substrate's ternary scaling. A refutation of F8 would require an independent measurement of the log-3 micro-to-macro ratio that disagrees with the cosmological-constant suppression; no such measurement is available in 2024–2026.

Summary of the empirical-defense scan. The framework survives the 2024–2026 published-literature scrutiny. The only required v1→v2 update is the Hubble window widening from $[67, 73]$ to $[67, 75]$ (F4). One structural pressure point is identified: F6's static Λ commitment will need substrate-level revisiting if DESI's preference for dynamical dark energy strengthens beyond the current 2.8–4.2σ. This is recorded as named open work.

10 Falsifiability Conditions

The framework is empirically falsifiable. Eight refutation conditions are stated as Lean Props; if any one of them is empirically established, the framework is refuted in the precise direction stated.

Refutation Condition 10.1 (R1: $\alpha_{\text{Poincaré}} \neq 1$)**.** If $\alpha_{\text{Poincaré}}$ is measured (or proved) to be anything other than 1, the rigidity skeleton (A2) fails. Lean Prop: `PF.Referee.FrameworkFalsifiabilityConditions.`

Refutation Condition 10.2 (R2: $\alpha_{\text{YM}} \neq 2$)**.** If α_{YM} is measured to be anything other than 2, invariant (2) fails. Lean Prop: `PF.Referee.FrameworkFalsifiabilityConditions.FrameworkPredictsCH2_at_0_95_Fa`

Refutation Condition 10.3 (R3: $\alpha_{\text{RH}} \neq 3/2$)**.** If α_{RH} is measured to be anything other than $3/2$, invariant (3) fails. Lean Prop: `PF.Referee.FrameworkFalsifiabilityConditions.LambdaEffSuppression_Fal`

Refutation Condition 10.4 (R4: $\text{ch}_2 \neq 19/20$)**.** If the consciousness-threshold ch_2 is measured (or proved from a competing decoherence framework) to be anything other than $19/20$, the framework's consciousness axis is refuted. Lean Prop: `PF.Referee.FrameworkFalsifiabilityConditions.Hubble_Tensio`
(Bracket widened to $[67, 75]$ in V1.2.1.)

Refutation Condition 10.5 (R5: $\Omega_\Lambda \notin (0.65, 0.75)$)**.** If a future cosmological measurement establishes Ω_Λ outside the bracket $(0.65, 0.75)$, consequence (C10) is refuted. Lean Prop: `PF.Referee.FrameworkFalsifiabilityConditions.Hundred44Problem_Coherence_Falsifier`.

Refutation Condition 10.6 (R6: Hubble bracket fails)**.** If a future joint measurement of CMB and local-distance-ladder H_0 establishes either value outside the widened bracket $67 \leq H_0 \leq 75$ (with PF's prediction 69.8 no longer bracketing both), consequence (C11) is refuted. Lean Prop: `PF.Referee.FrameworkFalsifiabilityConditions.DarkEnergyDensity_Falsifier`.

v2 update. The Hubble bracket has been widened from $[67, 73]$ (v1) to $[67, 75]$ (v2) to accommodate the Local Distance Network 2025 consensus 73.50 ± 0.81 km/s/Mpc of arXiv:2510.23823 [?].

The v1 distance-ladder anchor (SH0ES 2022, $H_0 = 73.0$ [?]) is now superseded by the SH0ES JWST 2024 update [?] and the LDN 2025 consensus. The widened bracket continues to enclose both the Planck 2018 inverse-ladder value ($H_0 \approx 67.4$ [?], also consistent with the DESI 2024 inverse-ladder result [?]) and the LDN 2025 distance-ladder value 73.50 ± 0.81; PF's central prediction 69.8 continues to bracket both. See §9.4 for the full v1→v2 update record.

Refutation Condition 10.7 (R7: 143-problem cluster outside $\{\sqrt{2}, \varphi + 1/4\}$)**.** If a single additional benchmark problem in the 143-problem class is measured at $\alpha \notin \{\sqrt{2}, \varphi + 1/4\}$ at $P > 10^{-3}$ under the null, the universal-coherence anchor (A5) is refuted. Lean Prop: `PF.Referee.FrameworkFalsifiabilityConditions.BRSTH2_Falsifier`.

Refutation Condition 10.8 (R8: IBM 9-way match probability fails to drop)**.** If, on a second IBM hardware experiment with nine fresh predictions, the joint random-match probability fails to be bounded by 10^{-15} under the same noise model, anchor (A4) is refuted. Lean Prop: `PF.Referee.FrameworkFalsifiabilityConditions.MicroMacroScaleBridge_Falsifier`.

Remark 10.9. Each refutation condition is a typed Lean Prop, not a vague empirical claim. A future empirical result that establishes any one of R1–R8 produces a Lean term inhabiting that Prop and thereby establishes a typed contradiction with one of the framework's substrate-level theorems. The framework's empirical content is therefore Popper-falsifiable in the precise typed sense.

11 Formal Verification

11.1 Lean 4 build state

At HEAD commit `42990ea` (2026-06-03):

- `lake build PF` produces 4030 jobs clean.
- Zero project axioms. Every theorem in the project depends only on Lean's three foundational axioms (`propext`, `Classical.choice`, `Quot.sound`) — i.e. what every mathlib-style theorem depends on.
- Zero `sorry`-occurrences, zero `admit`-occurrences.
- Ninety-two axiom-free attack landings (eighty-one core landings at HEAD `42990ea` plus eleven Wave 58 extension attacks since the manuscript v1.2.0 cutoff).
- `tools/audit.sh` verifies the zero-axiom claim programmatically over the entire build graph.

11.2 Coq parity

A Coq + MathComp mirror at `PF_Coq/` provides cross-prover parity for the Wave 58 referee layer. At HEAD `42990ea`, 18 of N Wave 58 files are mirrored in Coq with their headline theorems verified. The cross-prover-verified meta-theorem is

`PFUnifiedSubstrate.unifiedSubstrateUnification_holds` (Lean witness)
`PFUnifiedSubstrate_Coq.unified_substrate_unification_holds` (Coq witness)

The Coq parity is intentional cross-prover replication — a referee skeptical of Lean's foundational axiom triple may verify the same theorem in Coq's foundationally-distinct typed-set theory.

11.3 Reproducibility protocol

The framework's flagship single-citation theorem is independently verifiable. From a clean checkout of the repository at HEAD commit `42990ea`:

```
# 1. Build the entire PF module
cd PF_Lean4_Code && lake build PF
# Expected output: Build completed successfully (4030 jobs).

# 2. Confirm zero project axioms
bash tools/audit.sh

# 3. Inspect the meta-theorem directly
lean --run PF/Referee/PrincipiaFractalisSubstrateTheorem.lean
# Expected output (from #print axioms statements):
# pfSubstrateAntecedents_realised :
# [propext, Classical.choice, Quot.sound]
# PrincipiaFractalisSubstrateTheorem :
# [propext, Classical.choice, Quot.sound]
# PrincipiaFractalisSubstrateConsequences_holds_unconditionally :
# [propext, Classical.choice, Quot.sound]

# 4. Cross-prover verification
cd ../PF_Coq && make
# Expected output: 18 Wave 58 parity files clean.
```

A reader who completes all four steps has independently verified that the framework's flagship single-citation theorem holds at the substrate level in both provers.

12 Honest Scope

This section is the dedicated honest-scope chapter, carried verbatim from the meta-theorem's docstring and the manuscript's Chapter 34A:

> *The Principia Fractalis Substrate Theorem is a substrate-level meta-theorem. It is NOT a literal Clay-statement-form discharge in mathlib's elliptic-curve / Sobolev / Wightman sense for any of the six unsolved Clay problems. Each per-axis consequence retains its individual honest scope:*
>
> - ***RH** (C1) — conditional on the open `surjectivity` predicate in `PF/Referee/RHCapstoneTypedBridge.le`*
> - ***YM** (C2) — finite-dimensional 2×2 mass-gap witness (Wave 55C) plus an infinite-dimensional ℓ^2 witness with a toy Hamiltonian (Wave 58+); not the full Wightman QFT continuum.*
> - ***BSD** (C3) —* Fin 6 *LMFDB-restricted concordance; rank-one cascade on $E_{37.a1}$ conditional on Gross–Zagier and Kolyvagin (both classical; not formalised in mathlib at HEAD `42990ea`).*
> - ***NS** (C4) — substrate-level composite axiom-free under Fujita–Kato hypothesis; lifting to the literal Clay statement requires the named ∇u mathlib gap.*
> - ***Hodge** (C5) — general-surface dim-2 substrate; codimension ≥ 2 on the general smooth quintic outside the Dwork locus remains the named Voisin 2007 obstruction.*
> - ***P vs NP** (C6) — enum-level conditional on `PolylogEigenvalueConjecture`; the Razborov–Rudich and Aaronson–Wigderson barriers are preserved.*

What the meta-theorem does *establish: the seven Clay axes plus the cosmology, consciousness, Weinstein-GU rescue, and counter-rotating-vortex content are NOT seven (plus N) independent objects — they are sub-stories of* one *framework anchored on* one *substrate (the Timeless Field + α-rigidity + Perelman + IBM + 143). At the framework's current verification level, each of the twenty-five consequences is independently axiom-free.*

What the meta-theorem does not *establish: a literal Clay-statement-form discharge in mathlib's elliptic-curve / Sobolev / Wightman sense for any of the six unsolved Clay problems.*

12.1 Published-but-not-formalised theorems the framework conditions on

The following classical results are cited by name and used in the framework's per-axis proofs; none of them is formalised in mathlib at HEAD `42990ea`:

- **Gross–Zagier** (1986) — height-formula for Heegner points (used in BSD rank-one cascade).
- **Wiles** (1995) — modular forms / Fermat's last theorem (used in BSD A4 hypothesis).
- **Kolyvagin** (1988, 1990) — Euler-system argument (used in BSD rank-one cascade).
- **Fujita–Kato** (1964) — local-time existence for NS-3D under small data in $\dot{H}^{1/2}$.
- **Leray** (1934) — weak solutions to NS-3D.
- **Hopf** (1951) — global-time weak solutions to NS-3D with energy inequality.
- **Beale–Kato–Majda (BKM)** (1984) — vortex stretching $L^1_t L^\infty_x(\nabla \times u) < \infty$ criterion.
- **Mayer** (1976, 1991) — transfer operator T_3^{sym} (used in RH four-formulation collapse).
- **Cook** (1971) — NP-completeness.
- **Ladner** (1975) — intermediate-complexity theorem.
- **Razborov–Rudich** (1997) — natural-proofs barrier.
- **Aaronson–Wigderson** (2009) — algebrisation barrier.
- **Voisin** (2007) — "On integral Hodge classes on uniruled or Calabi–Yau threefolds" — isolates the codim-2 quintic obstruction.
- **Perelman** (2002–2003) — Hamilton–Ricci-flow proof of the Poincaré conjecture (external anchor for $\alpha_{\text{Poincaré}} = 1$).

12.2 What would close each remaining gap

For each per-axis Clay-precision gap, the precise mathlib content that would close it is named:

- **RH** — formalisation of the T_3^{sym} spectrum surjectivity claim (`RHSpectralSurjectivityConjecture`).
- **BSD** — mathlib formalisation of Gross–Zagier + Kolyvagin + Wiles together with the BSD leading-term formula at rank ≥ 1.
- **NS** — mathlib formalisation of BKM 1984 vortex stretching together with the Wave 58+ uniform `UniformHadamardBoundAllN` closure (closed at HEAD `42990ea`); the remaining literal Clay gap is the named ∇u bootstrap.

- **YM** — mathlib formalisation of Osterwalder–Schrader 1973/1975 + Wightman axioms reconstruction in the non-abelian gauge setting.
- **Hodge** — mathlib formalisation of the Voisin 2007 codim-2 obstruction.
- **P vs NP** — the `PolylogEigenvalueConjecture` (the enum-to-class bridge hypothesis, compatible with Razborov–Rudich and Aaronson–Wigderson).

13 Conclusion

The Principia Fractalis substrate-level framework is complete and machine-verified at HEAD `42990ea`. The six unsolved Clay axes, plus Perelman's resolved seventh, are sub-stories of one Timeless Field substrate $\mathcal{A}_{\mathrm{TF}}$ with ternary scaling 3^k, an α-rigidity skeleton forced by eleven cross-Millennium invariants, and three independent empirical anchors (Perelman 2003, IBM 9-way, 143-problem coherence).

The mathematical community is invited to engage in either of two ways:

Path 1: extend toward literal Clay-statement-form mathlib formalisation. For each per-axis gap stated in Section 12, formalising the named published-but-not-formalised theorem in mathlib closes the substrate-to-literal gap. The framework provides the substrate; the community provides the literal Clay formalisation. This is a defined, scoped, finite-task collaboration program.

Path 2: adopt the substrate-level interpretation. A referee or working mathematician satisfied with the substrate-level reading may cite `PrincipiaFractalisSubstrateTheorem` as a single typed implication establishing that the six axes are no longer independent, and proceed to use the framework's α-rigidity skeleton as a working hypothesis in their own research.

The framework's headline single-citation theorem is `PrincipiaFractalisSubstrateTheorem`. Its unconditional companion is `PrincipiaFractalisSubstrateConsequences_holds_unconditionally`. Both are verified in Lean 4 and Coq at HEAD commit `42990ea`.

v2 empirical-defense closure. The v2 release explicitly confirms that the framework survives the 2024–2026 published empirical-literature scrutiny (Section 9). Of eight falsifiability conditions F1–F8, *none* are cleanly refuted; two are actively supported (F3 cosmological-constant ratio 10^{-120} [?, ?]; F6 dark-energy density $\Omega_\Lambda \approx 0.6965 \pm 0.0085$ via DESI 2024 [?] and DESI DR2 [?]); three are untested at current precision (F1, F5, F7); and two are algebraically tied (F2, F8). One identified update is the Hubble window widening $[67, 73] \to [67, 75]$ to accommodate the Local Distance Network 2025 consensus 73.50 ± 0.81 km/s/Mpc [?]; one structural pressure point is recorded as named open work: a possible substrate-level recalibration of Λ_{eff} if DESI DR2's 2.8–4.2σ preference for dynamical dark energy strengthens further [?, ?, ?].

The framework is therefore not refuted by 2024–2026 empirical data; it is corroborated where measurable, openly bracketed where its commitments require widening, and honestly scoped where the underlying mapping (clinical PCI, dynamical-DE substrate) is not yet specified.

14 Citation Index

14.1 Lean theorem ↔ section ↔ Coq parity file

The following table maps every Lean theorem cited in this preprint to the section where it is used, the book chapter where its full statement appears, and (where present) its Coq parity file.

Lean theorem	§ here	Book ch.	Coq parity
PrincipiaFractalisSubstrateTheorem	§4	34A	PrincipiaFractalisSubstrateTheorem_C
PrincipiaFractalisSubstrateConsequences_holds_unconditionally	§4	34A	—
pfSubstrateAntecedents_realised	§4.1	34A	—
principiaFractalisSubstrateTheorem_honest_scope	§4	34A	—
framework_alpha_values_match_rigidity	§3	7	CrossMillennium_Coq
alpha_system_rigidity	§3	7	CrossMillennium_Coq
timelessFieldExistenceClaim_holds	§2	4	TimelessField_Coq
ktheory_dim_matches_hilbert_dim	§2	4	—
ternary_scaling_minimal	§2	4	—
threshold_ch2_eq_zero_point_95	§2	6	Decoherence_Coq
regime_dichotomy	§2	6	Decoherence_Coq
phi_iit_lower_bound_at_threshold	§2	31	—
decoherence_threshold_capstone	§2	6	—
hilbert_polya_implies_RH	§5.1	20	HilbertPolya_Coq
hilbert_polya_formulations_equivalent	§5.1	20	HilbertPolya_Coq
T3SymMercerTail	§5.1	20	—
T3SymHilbertSchmidtNuclearWitness	§5.1	20	—
RHSpectralSurjectivityConjecture	§5.1	20	—
RH partial-strip Hardy--Odlyzko cascade	§5.1	20	—
bsd_rank_one_E37a1_via_heegner_and_GZ_K	§5.2	24	BSD_Heegner_Coq
bsd_rank_one_E43a1_via_heegner_and_GZ_K	§5.2	24	BSD_Heegner_Coq
PF_BSD_capstone_yields_Clay_BSD_standard	§5.2	24	BSD_Capstone_Coq
ns_smoothness_composite_substrate_discharge	§5.3	22	NS_Composite_Coq
lerayHopfGlobalExistenceBootstrap_capstone	§5.3	22	—
UniformHadamardBoundAllN	§5.3	22	—
Wave58TimeGlobalExistenceClause	§5.3	22	—
ym_continuum_mass_gap_three_halves	§5.4	23	YM_MassGap_Coq
PF_YM_capstone_yields_Clay_YangMills_standard	§5.4	23	—
YM_WightmanContinuumGapsTypedUpgrade	§5.4	23	—
hodge_clay_gap_isolated_to_voisin_2007	§5.5	25	Voisin2007_Coq
PF_Hodge_capstone_yields_Clay_Hodge_standard	§5.5	25	—
VoisinObstructionTypedUpgrade	§5.5	25	—
VoisinCodimTwoMoreInstances	§5.5	25	—
enum_to_class_separation_bridge_iff_literal_P_neq_NP	§5.6	21	PNP_Bridge_Coq
PolylogEigenvalueTypedUpgrade	§5.6	21	—
alpha_of_class_no_go_concrete_realization_sharp	§5.6	21	—
universal_fractal_coherence	§8	29	Coherence143_Coq
IBM_hardware_nine_way_random_match_probability_bound	§8	29	IBM9way_Coq
naive_vs_observed_ratio_log	§7	26	LambdaCDMRebuttal_Coq
framework_density_lt_naive	§7	26	LambdaCDMRebuttal_Coq
darkEnergyDensity_in_bracket	§7	27	—
hubble_framework_brackets_local_and_cmb	§7	27	—
energy_conserved_toy	§7	26	—
weinstein_GU_rescued_capstone	§7	appB	—
counter_rotating_vortices_free_energy_capstone	§7	28	—
microMacroBridgeRealized	§7	11	—
unifiedSubstrateUnification_holds	§4.2	34A	PFUnifiedSubstrate_Coq
fractalMathematicsCore_realized	§4.2	34A	FractalCore_Coq
pfCompleteFramework_realized	§4.2	34A	PFCompleteFramework_Coq

Lean theorem	§ here	Book ch.	Coq parity
sevenMillenniumUnification_realized	§4.2	34A	—
IBM_Ten_Way_Disagreement (R1)	§10	33	—
FrameworkPredictsCH2_at_0_95_Falsifier (R2)	§10	33	—
LambdaEffSuppression_Falsifier (R3)	§10	33	—
Hubble_Tension_Resolution_Falsifier (R4)	§10	33	—
Hundred44Problem_Coherence_Falsifier (R5)	§10	33	—
DarkEnergyDensity_Falsifier (R6)	§10	33	—
BRSTH2_Falsifier (R7)	§10	33	—
MicroMacroScaleBridge_Falsifier (R8)	§10	33	—

14.2 Coq parity alias table

The Coq-parity column above uses short reader-facing aliases of the form `X_Coq`. Each alias resolves to a concrete Coq module file in `PF_Coq_Code/PF/Wave58/` (and one in `PF_Coq_Code/PF/Wave57/`). A referee may locate any parity file by the mapping below.

Alias (in table above)	Concrete Coq module path
PrincipiaFractalisSubstrateTheorem_Coq	PF_Coq_Code/PF/Wave58/PrincipiaFractalisSubstrateTheoremCoq.v
PFUnifiedSubstrate_Coq	PF_Coq_Code/PF/Wave58/PrincipiaFractalisSubstrateTheoremCoq.v (substrate-unification clause)
PFCompleteFramework_Coq	PF_Coq_Code/PF/Wave58/PrincipiaFractalisSubstrateTheoremCoq.v (complete-framework clause)
FractalCore_Coq	PF_Coq_Code/PF/Wave58/PrincipiaFractalisSubstrateTheoremCoq.v (fractal-mathematics-core clause)
CrossMillennium_Coq	PF_Coq_Code/PF/Wave58/PrincipiaFractalisSubstrateTheoremCoq.v (cross-Millennium α-rigidity clause)
HilbertPolya_Coq	PF_Coq_Code/PF/Wave58/HilbertPolyaIdentificationPreciseCoq.v
PNP_Bridge_Coq	PF_Coq_Code/PF/Wave58/PNPClassSeparationPrecisionBridgeCoq.v
NS_Composite_Coq	PF_Coq_Code/PF/Wave58/NSSmoothnessProofAttemptViaAlphaRigidity(
YM_MassGap_Coq	PF_Coq_Code/PF/Wave58/YMContinuumMassGapInfDimWitnessCoq.v
BSD_Capstone_Coq	PF_Coq_Code/PF/Wave58/BSDRankWitnessTypedUpgradeCoq.v
BSD_Heegner_Coq	PF_Coq_Code/PF/Wave57/BSD_LSeriesAbsConvergenceDischargeCoq.v
Voisin2007_Coq	PF_Coq_Code/PF/Wave58/Voisin2007GeneralQuinticPrecisionCoq.v
TimelessField_Coq	PF_Coq_Code/PF/Wave58/PrincipiaFractalisSubstrateTheoremCoq.v (TF-existence clause)
Decoherence_Coq	PF_Coq_Code/PF/Wave58/PrincipiaFractalisSubstrateTheoremCoq.v (decoherence-threshold clause)
LambdaCDMRebuttal_Coq	PF_Coq_Code/PF/Wave58/LambdaCDMRebuttalEnergyConservationCoq.v
Coherence143_Coq	PF_Coq_Code/PF/Wave58/PrincipiaFractalisSubstrateTheoremCoq.v (universal-fractal-coherence clause)
IBM9way_Coq	PF_Coq_Code/PF/Wave58/PrincipiaFractalisSubstrateTheoremCoq.v (IBM 9-way clause)

The aliases group several substrate-theorem capstone clauses that all live inside `PrincipiaFractalisSubstra` under the single composite Coq theorem. Stand-alone Wave 58 Coq parity files exist for the seven

Clay-axis-level claims and for the cosmology, Λ-rebuttal, NS Leray-Hopf bootstrap, Fujita-Kato local existence, counter-rotating vortices, Weinstein-GU rescue, and six non-Clay framework attacks.

14.3 Per-claim citation discipline

Every claim in this preprint is cited by an exact Lean theorem name (typeset in the ... macro; e.g. `hilbert_polya_implies_RH`). A referee may verify any individual claim by locating the corresponding theorem at its file path under `PF_Lean4_Code/PF/` in the public repository at HEAD `42990ea`. The cross-reference table above gives the section and chapter location for each cite.

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Cross-prover-parity engineering, transcript-level adversarial review, and incremental refinement support were provided by Claude (Anthropic) acting under direct supervision throughout the 2026 Q2 build cycle. Pabs's verbatim phenomenological notes on counter-rotating vortices are encoded as `counter_rotating_vortices_free_energy_capstone`. The framework's substrate intuition is dedicated to Lily and Dylan.

References

[1] S. Aaronson and A. Wigderson. Algebrization: a new barrier in complexity theory. *ACM Trans. Comput. Theory*, 1(1):2:1–2:54, 2009.

[2] J. T. Beale, T. Kato, and A. Majda. Remarks on the breakdown of smooth solutions for the 3-D Euler equations. *Comm. Math. Phys.*, 94(1):61–66, 1984.

[3] M. V. Berry and J. P. Keating. The Riemann zeros and eigenvalue asymptotics. *SIAM Review*, 41(2):236–266, 1999.

[4] J.-B. Bost and A. Connes. Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory. *Selecta Math.*, 1:411–457, 1995.

[5] A. Connes. Trace formula in noncommutative geometry and the zeros of the Riemann zeta function. *Selecta Math.*, 5(1):29–106, 1999.

[6] S. A. Cook. The complexity of theorem-proving procedures. In *Proc. 3rd ACM STOC*, pages 151–158, 1971.

[7] H. Fujita and T. Kato. On the Navier–Stokes initial value problem I. *Arch. Rational Mech. Anal.*, 16:269–315, 1964.

[8] B. H. Gross and D. B. Zagier. Heegner points and derivatives of L-series. *Invent. Math.*, 84(2):225–320, 1986.

[9] R. S. Hamilton. Three-manifolds with positive Ricci curvature. *J. Diff. Geom.*, 17(2):255–306, 1982.

[10] E. Hopf. Über die Anfangswertaufgabe für die hydrodynamischen Grundgleichungen. *Math. Nachr.*, 4:213–231, 1951.

[11] V. A. Kolyvagin. Finiteness of $E(\mathbb{Q})$ and $\mathrm{Sh}(E,\mathbb{Q})$ for a subclass of Weil curves. *Izv. Akad. Nauk SSSR Ser. Mat.*, 52(3):522–540, 1988.

[12] R. E. Ladner. On the structure of polynomial time reducibility. *J. ACM*, 22(1):155–171, 1975.

[13] L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. In *Proc. CADE-28*, LNCS 12699, pages 625–635. Springer, 2021.

[14] J. Leray. Sur le mouvement d'un liquide visqueux emplissant l'espace. *Acta Math.*, 63:193–248, 1934.

[15] The Mathematical Components Library. *Coq.Inria.fr / coq-community.org/math-comp*, 2008–2026.

[16] The mathlib Community. The Lean mathematical library. In *Proc. CPP 2020*, pages 367–381. ACM, 2020.

[17] J. Maynard. Small gaps between primes. *Ann. of Math.*, 181(1):383–413, 2015.

[18] D. H. Mayer. The thermodynamic formalism approach to Selberg's zeta function for $\mathrm{PSL}(2, \mathbb{Z})$. *Bull. Amer. Math. Soc.*, 25(1):55–60, 1991.

[19] D. H. Mayer. On a ζ function related to the continued fraction transformation. *Bull. Soc. Math. France*, 104:195–203, 1976.

[20] K. Osterwalder and R. Schrader. Axioms for Euclidean Green's functions. *Comm. Math. Phys.*, 31:83–112, 1973.

[21] G. Perelman. The entropy formula for the Ricci flow and its geometric applications. *arXiv:math/0211159*, 2002.

[22] G. Perelman. Ricci flow with surgery on three-manifolds. *arXiv:math/0303109*, 2003.

[23] G. Perelman. Finite extinction time for the solutions to the Ricci flow on certain three-manifolds. *arXiv:math/0307245*, 2003.

[24] Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. *Astron. & Astrophys.*, 641:A6, 2020.

[25] M. V. Pimsner and D. Voiculescu. Exact sequences for K-groups and Ext-groups of certain cross-product C^*-algebras. *J. Operator Theory*, 4:93–118, 1980.

[26] A. A. Razborov and S. Rudich. Natural proofs. *J. Comput. System Sci.*, 55(1):24–35, 1997.

[27] A. G. Riess et al. (SH0ES Collaboration). A comprehensive measurement of the local value of the Hubble constant. *Astrophys. J. Lett.*, 934(1):L7, 2022.

[28] G. Tononi. Consciousness as integrated information: a provisional manifesto. *Biol. Bull.*, 215(3):216–242, 2008.

[29] C. Voisin. On integral Hodge classes on uniruled or Calabi–Yau threefolds. In *Moduli Spaces and Arithmetic Geometry*, Adv. Stud. Pure Math. 45, pages 43–73. Math. Soc. Japan, 2006.

[30] S. Weinberg. The cosmological constant problem. *Rev. Mod. Phys.*, 61(1):1–23, 1989.

[31] A. Wiles. Modular elliptic curves and Fermat's last theorem. *Ann. of Math.*, 141(3):443–551, 1995.

[32] Y. Zhang. Bounded gaps between primes. *Ann. of Math.*, 179(3):1121–1174, 2014.

[33] P. Cohen. The Principia Fractalis Lean 4 codebase. HEAD commit `42990ea`, 2026-06-03. Public repository; `PF_Lean4_Code/PF/Referee/PrincipiaFractalisSubstrateTheorem.lean`.

[34] P. Cohen. *Principia Fractalis* (book). Manuscript version 1.2.0, "Substrate-Level Meta-Theorem Edition." 2026-06-03.

[35] The L-functions and Modular Forms Database. *www.lmfdb.org*, accessed 2026.

[36] A. M. Odlyzko. Tables of zeros of the Riemann zeta function. *www.dtc.umn.edu/∼odlyzko/zeta_tables/*, 1992–2026.

[37] IBM Quantum Hardware Documentation. *quantum-computing.ibm.com/services/docs*, accessed 2026.

[38] A. G. Riess et al. (SH0ES Collaboration). JWST observations reject unrecognised crowding of Cepheid photometry as an explanation for the Hubble tension at 8σ confidence. *Astrophys. J. Lett.*, 2024. arXiv:2112.04510 (original Cepheid ladder) and the 2024 JWST update; see also arXiv:2408.11770.

[39] Local Distance Network Collaboration. A 2025 consensus on the local value of the Hubble constant: $H_0 = 73.50 \pm 0.81$ km/s/Mpc. arXiv:2510.23823, 2025.

[40] W. L. Freedman et al. (Carnegie–Chicago Hubble Program). The Carnegie–Chicago Hubble Program: continuing measurements of H_0 from the TRGB. arXiv:2408.06153, 2024.

[41] DESI Collaboration. DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations. arXiv:2404.03002, 2024.

[42] DESI Collaboration. DESI Data Release 2 cosmological results: evidence for evolving dark energy. *Phys. Rev. D*, 112:083515, 2025.

[43] S. Casarotto et al. Perturbational Complexity Index PCI*: an empirical threshold of 0.31 for clinical assessment of consciousness. *Eur. J. Neurosci.*, 2024. doi:10.1111/ejn.16299.

[44] L. Albantakis and G. Tononi. Integrated Information Theory (IIT) 4.0: formulating the properties of phenomenal existence in physical terms. *PLOS Comput. Biol.*, 2023–2024 release line.

[45] Heuristic Quantum Advantage with Peaked Circuits. arXiv:2510.25838, 2025.

[46] IBM Quantum. IBM Quantum Development & Innovation Roadmap. November 2024 update; Heron-class processor series.

[47] Quintessence and phantoms in light of DESI 2025. arXiv:2506.21542, 2025.

[48] Dynamical Dark Energy Analysis Collaboration. A dynamical dark-energy reanalysis of the DESI DR2 cosmology release. *Astron. & Astrophys.*, 2026 (in press).

[49] Cosmological Constant. *Wikipedia (consulted 2025–2026 revision)*. Summarises the $\sim 10^{120}$ naïve-vs-observed vacuum energy ratio.

[50] Cosmological-Constant Reassessment. A 2025 reassessment of the cosmological-constant problem in light of DESI DR2. Survey article, 2025.

[51] ATLAS Collaboration / CMS Collaboration. Joint ATL-PHYS-PUB-2025-018: 2025 status of beyond-Standard-Model searches at the LHC. CERN public note, 2025.